
Impact of Advanced AI Agents on the Enforcement-Circumvention Balance in Export Controls

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Abstract

U.S. export controls on advanced AI chips have raised acquisition costs but have not prevented large-scale diversion to restricted destinations. Recent DOJ charges allege over \$670M in diverted AI hardware across just two cases. As AI agents become capable of multi-step autonomous task execution, both enforcement and circumvention workflows become candidates for automation. We distinguish informational constraints, whose binding force derives from the cost and speed of processing information, from structural constraints, whose relaxation requires changes in legal authority, physical presence, or jurisdictional reach. AI agents relax informational constraints on both sides, but do so unevenly: the binding constraints on circumvention are disproportionately informational, while the binding constraints on enforcement are disproportionately structural. We map seven agentic capability domains onto paired enforcement and circumvention workflows, and identify three observable indicators that distinguish which adoption trajectory is materializing. Under default adoption trajectories, AI-enhanced screening scales faster than the institutional mechanisms that convert flags into holds, denials, or seizures, widening a detect-to-action gap that favors circumvention. Without reforms linking detection to action at operational chokepoints, expanded AI tooling will increase enforcement workload without proportionally reducing diversion.

1. Introduction

U.S. export controls on advanced AI chips are a primary instrument of national security strategy, intended to slow adversary access to frontier AI capabilities. The Bureau

of Industry and Security (BIS) administers the Export Administration Regulations (EAR), which govern the export, reexport, and in-country transfer of dual-use technologies, including advanced semiconductors and AI accelerators (Bureau of Industry and Security, 2022; Heim, 2025). These controls rely on licensing requirements, end-use restrictions, and post-shipment compliance mechanisms to constrain access to high-performance computing, increasingly viewed as a prerequisite for national technological advantage.

Export controls have made adversary compute acquisition more expensive, with black-market prices for advanced AI chips reportedly running roughly 50% above U.S. list prices (DIGITIMES, 2025; Financial Times, 2025). Enforcement capacity has not expanded at the same rate. BIS has requested over \$100 million in additional funding to support licensing review and post-shipment verification (Government Accountability Office, 2025). Diversion activity has continued in parallel. Recent DOJ cases allege more than \$670 million in diverted AI hardware across two prosecutions, including at least \$160 million in controlled NVIDIA H100 and H200 GPUs in Operation Gatekeeper and approximately \$510 million in U.S.-assembled AI servers in a separate March 2026 indictment (Department of Justice, 2025; U.S. Department of Justice, 2026). Trade-press reporting based on interviews with network participants describes at least eight Chinese procurement networks of comparable scale (Liu, 2024); no official estimate has been published.

Autonomous agents now run discovery tasks at large law firms, due-diligence workflows at supply-chain risk vendors such as Altana and Sayari, and software engineering tasks at frontier labs. The same task profiles appear on both sides of the enforcement divide: reading documents, resolving entities across registries, and planning multi-step actions. On GAIA, the top verified score on the public validation set reached 74.55 percent in September 2025 (Mialon et al., 2024) and 79 percent on SWE-bench Verified (Scale AI, 2026), up from 15 and 33 percent respectively in mid-2024, and the time horizon of reliably completed tasks is doubling roughly every seven months (Kwa et al., 2025). Entity resolution across corporate registries, currently an investigator-weeks task, is becoming a model-minutes task, and the same capability lowers the cost for circumvention networks producing front-company documentation at scale.

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Main contributions. We analyze how agentic AI changes the enforcement-circumvention contest in export controls, using a structured comparison of what agentic capabilities can and cannot relax on each side. To provide structural frameworks for our analysis, we leverage the offense-defense literature in international security (Jervis, 1978; Glaser & Kaufmann, 1998; Van Evera, 1998) and its adaptation to cybersecurity (Slayton, 2017; Garfinkel & Dafoe, 2019). The cybersecurity analogy is useful but limited: export-control contests turn on reconstitution speed and detection latency rather than vulnerability surface area. For example, a successful security exploit can be repeated until the vulnerability is patched, while in export controls, a successful diversion permanently transfers physical goods, but the network that executed it accumulates exposure with each transaction and can be disrupted through prosecution. Conversely, patching a software vulnerability protects all systems simultaneously, whereas interdicting one smuggling network does nothing to prevent the next. The variables that matter in export controls are therefore reconstitution speed, how quickly a disrupted network can rebuild, and detection latency, how many successful diversions occur before a network is identified, rather than vulnerability surface area as conventionally understood in cybersecurity.

Our analysis draws on DOJ prosecution materials, BIS enforcement actions, GAO audits, AI capability benchmarks, and seven semi-structured background interviews conducted between January and March 2026: three with individuals with BIS or BIS-adjacent export-control experience, three with researchers working on compute governance, and one with an industry practitioner at a major AI accelerator firm. We use these interviews to validate workflow assumptions, identify operational bottlenecks, and check whether the constraint classifications reflected practitioner experience. We do not treat the interviews as a representative sample or as independent empirical measurement. The confidence scores in Table 6 and the constraint decomposition in Tables 4 and 5 are structured qualitative judgments, produced through independent scoring by each author and reconciled through discussion. We report these as judgments, not measurements, and specify disconfirmation criteria in §G.

- We distinguish informational constraints that AI can relax (screening, triage, documentation) from structural constraints that remain binding regardless of computational capability (physical verification, legal authority, jurisdictional reach), and show that the constraints AI relaxes on the circumvention side are closer to current operational limits than those it relaxes on the enforcement side.
- We assess seven agentic AI capability domains and map their differential impact onto enforcement and circumvention task chains using benchmark evidence and enforcement-action records.
- We sketch two stylized 2027–2029 scenarios that make

the capability assessments concrete, and identify three observable indicators—diversion success rate, detect-to-hold conversion speed, and the share of diversion shifting to post-shipment channels—that would signal which trajectory is materializing.

2. Background

Export control enforcement can fail in two distinct ways. It can fail to identify suspicious transactions because data is fragmented, entities are difficult to resolve across jurisdictions, or screening capacity is thin. Or it can identify suspicious transactions and still fail to stop them, because physical verification is scarce, jurisdictional authority weakens after the first export hop, and institutional handoffs are slow. These are different failure modes that respond to different interventions. Sections 3 through 5 then use that map to assess how AI adoption changes the balance.

2.1. The Enforcement Architecture

Government. U.S. export enforcement is centered in the Bureau of Industry and Security (BIS), operating through a chain of linked functions rather than a single actor. BIS administers and interprets the Export Administration Regulations (EAR) through its rulemaking and licensing architecture. Specifically, its enforcement arm, the Office of Export Enforcement (OEE), investigates suspected violations—diversion networks, falsified documentation, and illicit procurement. Importantly, the staff resourcing structure means that the system can review and flag far more suspicious cases than it can physically investigate.

The BIS also relies on two main verification tools. Pre-License Checks (PLCs) assess whether a proposed buyer and end use appear credible before a license is granted. Post-Shipment Verifications (PSVs) attempt to confirm that exported items are where they were declared to be and used as represented. Both tools are valuable, but they are scarce and slow relative to the volume of global controlled-goods trade. Operationally, enforcement depends on a small number of structural chokepoints, e.g., the original manufacturer or designer, the distribution layer of authorized distributors, and the downstream compute layer (Heim, 2025). Ease of control varies by item type. Semiconductor manufacturing equipment is visible and service-intensive. Commodity chips are hard to police because they are low-value and high-volume. Advanced AI accelerators occupy the hardest position: compact, extremely high-value, and often untraceable once they enter reseller or cloud supply chains. This creates a distinction between visibility at the first export and control after resale. AI can improve analysis of the initial transaction record, but it cannot reconstruct downstream ownership or effective use when later transfers are not reported.

Exporter Compliance. Export controls depend heavily on

private-sector compliance. More precisely, large upstream firms operate structured compliance programs that include customer verification, end-user certificates, restricted-party screening, internal reviews, and license applications where required. Two features of private compliance shape its practical limits. First, compliance expectations are framed through guidance and “reasonable” due diligence rather than tightly specified operational requirements, leaving firms significant discretion in how they implement screening and escalation (Supplement No. 3 to Part 732). Second, diligence consistently weakens downstream, where large upstream firms typically have stronger controls and better data access than entities further along the chain. As a result, compliance programs are often designed to manage legal liability in ordinary transactions rather than to detect sophisticated front-company procurement or multi-hop diversion networks—a distinction that matters when adversaries deliberately exploit the weaker segments of the distribution chain. AI-assisted compliance is therefore likely to be strongest at the upstream firms that are already easiest to monitor. Circumvention networks can respond by shifting activity toward smaller resellers, integrators, and logistics intermediaries where data and incentives are thinner.

Shared Constraints. Taken together, government enforcement and private compliance form a system that is better at accumulating evidence of risk than at converting that evidence into interrupted diversion. Constraints on finding risky cases include fragmented data across siloed systems, misclassification and misreporting risk in exporter filings, and weak entity resolution across jurisdictions (Center for Strategic and International Studies, 2022). In contrast, constraints on verifying and stopping risky cases require physical presence—limiting scalability. Moreover, downstream visibility follows a gradient: strong at the initial U.S. export, weak at the first foreign resale, and often nonexistent by the second or third transfer (Heim, 2025). Even where serial numbers exist, end-to-end traceability across resale, server integration, and data center deployment is often incomplete, producing a persistent gap in which suspicion rises faster than verifiable location can be established. Jurisdictional barriers compound this: overseas end-use checks are often voluntary, and may necessitate inter-agency handoffs across licensing, customs, investigation, and prosecution (Government Accountability Office, 2025; Center for Strategic and International Studies, 2022). We provide a summary of these constraints in Table 3.

3. Circumvention Pathways, Scale, and Operational Bottlenecks

Unlike resource-constrained enforcement agencies, circumvention networks benefit from strong financial incentives created by black-market price premiums and, in some cases, demand linked to state-directed technology priorities. In

this section, we map the dominant diversion pathways, assess how agentic AI affects each, and identify the operational bottlenecks that currently constrain circumvention, distinguishing those that AI can relax from those that remain structurally binding.

3.1. Dominant Diversion Pathways

Straw-buyer networks. Smuggling networks often utilize U.S.-registered entities posing as legitimate domestic end users to procure chips from U.S. distributors. Chips are aggregated in domestic warehouses where workers physically remove NVIDIA serial number labels and branding, then relabel them as fictitious brands and describe them as “generic computer parts” (Department of Justice, 2025), complicating tracking of origin and destination.

Phantom data centers. Shell companies in transshipment hub jurisdictions such as Malaysia, Thailand, Vietnam, and the UAE purchase full server racks ostensibly for local cloud provision. Facilities are built to specification to satisfy initial inspections. Following the inspector’s departure, centers are liquidated, and racks are disassembled. The suitcase method for Blackwell-series GPUs involves disassembling server chassis into individual GPU modules worth tens of thousands of dollars each, then transporting them via land or short-haul flights into southern China, bypassing maritime cargo screenings. Singapore accounts for 18 to 22 percent of NVIDIA revenue despite representing less than 2 percent of physically shipped product volume, a discrepancy consistent with significant transshipment activity (CNBC, 2025). NVIDIA’s own 10-Q filing discloses that over 99 percent of controlled Data Center compute revenue billed to Singapore was for orders from U.S.-based customers (NVIDIA Corporation, 2025), suggesting the revenue-share figure reflects billing geography rather than end-use demand.

Shell company proliferation. The scale of shell company use in sanctions and export control evasion is well documented across contexts. It is estimated that \$4 billion in export-controlled semiconductor chips have reached Russia since 2022 through more than 6,000 intermediary companies, many with minimal web presence, virtual office addresses, and no demonstrable real operations (New York Times, 2024). In addition, the KSE Institute analyses document similar patterns in Russian military procurement more broadly, including the use of nominee structures and opaque jurisdictions to circumvent screening (Bilousova et al., 2024). Similar structures characterize Chinese semiconductor procurement networks.

Black-market economics. Market prices sustain and incentivize circumvention. Nvidia B200 rack prices in China are reported at \$420,000 to \$490,000 versus \$280,000 to \$330,000 in the United States, representing an approximately 50% black-market premium (Financial Times, 2025; DIGITIMES, 2025). These margins sustain

continued investment in circumvention infrastructure.

3.2. How Agentic AI Affects Circumvention Pathways

AI agents can accelerate the most labor-intensive steps in building synthetic supply chains. Filing incorporation documents across multiple jurisdictions currently requires human operators to navigate each jurisdiction’s registry interface, prepare jurisdiction-specific documentation, and maintain consistency across filings. Agentic systems with multi-step task completion capabilities have demonstrated abilities to execute workflows autonomously (Simular, 2026). In circumvention, this can materialize as generating internally consistent sets of trade documents, bills of lading, invoices, and end-user certificates that vary enough to evade template-matching detection.

Physical transport remains irreducibly dependent on human operators, but the physical characteristics of the goods limit its constraining force. The suitcase method for Blackwell-series GPUs, disassembling server chassis and moving individual modules overland into southern China, shows that physical logistics function as an operational inconvenience rather than a chokepoint. As AI sharpens KYC screening on the enforcement side, circumventors will respond with higher-quality synthetic documentation, shifting the contest toward the thinnest enforcement capacities: physical inspections and post-shipment verifications, discussed in Section 2.1. The adoption threshold is also lower for circumvention. A model that is too unreliable for government deployment may still be useful for drafting filings, producing variant invoices, monitoring registries, or coordinating repetitive logistics tasks under human supervision.

But increased throughput does not give circumvention an unqualified advantage. Circumvention networks can retry cheaply when an incorporation filing is rejected or a forged document fails to convince. These are operational failures, and they cost almost nothing. Detection failures are catastrophic. A single flagged shipment can trigger prosecution, asset seizure, and the unraveling of connected intermediaries. AI-assisted circumvention generates synthetic entities and documentation at higher volume, but each additional entity and transaction also creates a signal that enforcement could find. The same tools that let a network spin up twenty shell companies in a week also produce twenty new nodes for graph-based anomaly detection to flag. Whether circumvention gains ground depends on how fast enforcement can work through that expanded signal surface relative to the rate at which new synthetic fronts appear.

3.3. Circumvention Bottlenecks

Contemporary circumvention remains constrained by operational bottlenecks that AI cannot easily relax.

Trusted human intermediaries. Circumvention continues to depend on trusted human intermediaries at multiple stages of the supply chain, e.g., procurement agents, warehouse operators, and logistics coordinators. While AI systems can assist with screening, scheduling, and communication, they cannot substitute for the trust relationships required to physically handle controlled goods, falsify records at points of custody, or coordinate across jurisdictions without triggering suspicion. As networks expand to increase throughput, the number of participants with operational visibility grows, raising the probability of failure.

Physical handling requirements. Export-controlled chips introduce physical constraints: they must be stored, repackaged, shipped, and installed in real-world facilities. Even when shipments are disguised or mis-declared, they must traverse ports, customs facilities, and data centers subject to inspection and post-hoc verification. These requirements limit the extent to which circumvention can be scaled.

Documentation coherence at scale. Documentation and identity fraud, while increasingly commoditized, introduce coordination constraints. Generating convincing end-user certificates, customs declarations, and corporate records at scale requires internal consistency across documents, entities, and transactions. As the number of shell companies and shipments grows, maintaining coherence becomes operationally complex. Errors or inconsistencies propagate across the network, increasing the likelihood that a single anomaly exposes broader patterns. AI-enabled throughput shifts the bottleneck from document quality to document coherence under scale, a constraint that graph-based anomaly detection on the enforcement side may be well-positioned to exploit.

Financial routing. Payments for high-value semiconductor transactions must ultimately move through regulated financial systems, often across multiple jurisdictions. Layering transactions to obscure end users increases operational complexity and latency, while concentrating payments raises detection risk. Operation Gatekeeper documented over \$50 million in direct wire transfers from the PRC, routed through intermediate jurisdictions to obscure their origin (Department of Justice, 2025). Even with automated tooling, mismatches between declared destinations, counterparties, and payment origins remain a recurring point of failure. Once goods or operational control move into China, legal access to internal records becomes a structural constraint rather than an analytic one. AI can improve inference from external signals, but it cannot compel production of records that foreign authorities cannot lawfully obtain.

Reconfiguration costs. While diversion networks can reorganize in response to new controls or enforcement actions, each cycle of reconstitution requires dissolving entities, rerouting logistics, and establishing new commercial histories. These reset cycles temporarily reduce throughput and degrade the trust relationships on which operations depend.

3.4. Informational vs. Structural Constraints

We classify constraints by whether AI can plausibly relax the mechanism that makes them binding. A constraint is informational and therefore tractable for AI if three conditions hold:

1. **Information-processing character:** Binding force derived primarily from the cost, speed, or accuracy of processing, linking, or generating information.
2. **Demonstrated or near-term AI applicability:** Existing benchmarks or deployed systems provide evidence that AI systems can perform the relevant task at or near human-level quality within the 2026–2029 window.
3. **No non-informational dependency:** Relaxing the constraint does not require changes in legal authority, physical presence, jurisdictional reach, institutional coordination mechanisms, or procurement timelines.

A constraint is classified as structurally binding if relaxing it requires action outside the domain of information processing—i.e., if any of the three criteria above is not met. The key implication is that even unlimited computational capability cannot relax a structurally binding constraint. Tables 5 and 4 provide examples of the constraint inventory.

Aggregate assessment. AI-relaxable enforcement constraints are genuine bottlenecks, but relaxing them mostly accelerates screening and triage rather than interdiction. Concretely, available evidence suggests AI can increase screening and triage throughput by a factor of two to three in the near term (Dell’Acqua et al., 2023; McKinsey & Company, 2025; Allen, 2025). We use the conservative end of available evidence—a throughput improvement of roughly two to three times for screening and triage tasks—while noting that higher gains are plausible for narrow subtasks such as document sorting or name matching, where near-full automation is feasible. In contrast, the stages where enforcement must convert flags into holds scale far more slowly, as they require a combination of legal authority, physical presence, and institutional coordination.

We further note that AI expands the front end of the enforcement pipeline without proportionally expanding the back end. Circumvention faces the opposite distribution: its tightest constraints, documentation coherence, multi-step coordination, and shell company creation, are informational, and current AI systems can address them directly. When AI relaxes these constraints, circumvention can scale faster, because the remaining structurally binding constraints impose per-transaction costs rather than hard limits on volume. Financial routing through regulated systems is the most significant remaining chokepoint. Payments create traceable signatures that AI cannot eliminate, though it can help circumventors layer and obfuscate transactions more efficiently.

Loss of visibility after initial export has both informational and structural components. Monitoring improves only

within the data that institutions already possess. It does not solve missing records or legal barriers to access. AI-enhanced monitoring is therefore likely to hit a ceiling without parallel expansion of reporting requirements and legal authority over post-shipment transfers. This constraint is central to the displacement dynamics analyzed in Section 5.

Weighting the inventory by importance does not yield a defensible ratio: weights were assigned by co-author judgment, and the inventory is unlikely to be exhaustive. What it yields is a direction. On the enforcement side, the heavily weighted constraints are mostly structural: physical inspection capacity, legal authority, jurisdictional reach, and inter-agency coordination. On the circumvention side, they are mostly informational: documentation coherence, multi-step coordination, and shell company creation. That direction does not depend on the weights being exactly right. Reversing it would require reclassifications the operational evidence rules out, either treating physical verification as an information-processing problem or treating documentation coherence as structurally binding. It is also stable under reasonable shifts within the current classification. Raising interagency coordination from medium to high, for example, narrows the gap without closing it; closing it would require several structural enforcement constraints to be reclassified at once, and the evidence in Sections F and G does not support any such reclassification.

4. Agentic Nearcast: How Capable AI Is Likely to Improve Offense–Defense Pathways

4.1. AI Agents as Enforcement/Circumvention Tools

We now assess the current and near-term trajectory of specific agentic capabilities and evaluate their likely impact on the enforcement–circumvention balance through 2028. The scope of our analysis is restricted to agentic capabilities that satisfy the following selection criteria:

- **Direct workflow relevance.** Capabilities that map to specific tasks documented in BIS enforcement actions, DOJ prosecutions, or think tank analyses.
- **Most transformative value.** Capabilities that we anticipate to change rapidly during the 2026–2028 window and remain a priority to frontier labs, inferred from benchmarks, expert forecasts, and broader deployment signals.

We construct an impact consensus that is measured using two confidence axes. First, we estimate confidence that agentic capability will have significantly evolved by 2028. Second, we measure confidence that the capability, once mature, can meaningfully impact the current dynamic. Our analysis is based on a systematic review of frontier AI lab capability roadmaps and benchmark suites (METR, SWEbench, WebArena) and a review of BIS enforcement actions and DOJ prosecutions to identify capability requirements for both enforcement and circumvention workflows. We con-

strain confidence range by the quality of available evidence: 75% and higher (Tier 1 evidence) corresponds to high-signal data from multiple expert/lab forecasts, benchmark trends, and deployed systems; 50–75% (Tier 2) corresponds to some expert views and benchmarks or direct deployment evidence; 50% and lower (Tier 3) primarily involves trend extrapolation, market signals, and speculative judgment.

Impact of agentic capabilities. In Table 6 we report the seven capability domains, the confidence assigned to each forecast, and the expected directional tilt. The three capabilities with the strongest asymmetries are Evidence Collection, Autonomous Engineering and Planning, and Cybersecurity. The confidence scale ranges from **5 (Decisive)** where multiple independent evidence lines converge to **1 (Minimal)**: Speculative with no direct evidence. Lastly, net impact reflects the direction of tilt under default adoption trajectories.

In appendix Section C.2, we discuss three major agentic capabilities in depth and assess their near-term impact on the evolution of enforcement-circumvention dynamics. Autonomous evidence-collection agents are already deployed at scale (Sayari, Altana) and favor enforcement, which has legitimate access to government databases that circumvention actors cannot reach. For the capabilities involving autonomous engineering and planning, we find that current frontier models can solve complex objectives in software engineering by formulating subtasks and autonomously writing, testing, and deploying software at scale. Acceleration in this capability favors circumvention on three margins: faster adoption, higher tolerance for execution failures, and lower coordination complexity, since circumventors plan against a slow, legally constrained opponent while enforcement plans against an opaque, adaptive one. Cybersecurity is the third asymmetry, and the most pronounced. Capabilities have advanced quickly on both sides, but the structural imbalance favors offense: circumvention needs one successful jailbreak or prompt injection, while enforcement must hold across the full attack surface of systems processing untrusted external data. Recent evaluations show frontier models approaching expert-level performance on autonomous vulnerability discovery and multi-stage attack simulation, with Anthropic reporting expert validators agreeing with model-generated severity assessments in 89 percent of reviewed vulnerability reports (Anthropic, 2026) and AISI finding that Claude Mythos Preview can autonomously execute multi-stage attacks on vulnerable networks (AI Security Institute, 2026). Defensive robustness continues to lag: twelve published prompt-injection defenses were bypassed at attack success rates above 90 percent (Souly et al., 2025). Together, these three asymmetries compound into the *detect-to-action* gap developed in Section C.3. The default trajectory assumes sophisticated networks adopt frontier capabilities; smaller or less technically equipped networks will lag, and our claims describe the upper bound of the circumvention

advantage rather than a uniform outcome.

To extend the framework beyond the seven capabilities assessed here, we pair enforcement workflows with their circumvention counterparts and identify the binding constraint at each pairing (Table 7, Section D).

4.2. Nearcasting via Stylized Example Scenarios

Two stylized scenarios illustrate how these capabilities enter enforcement and circumvention workflows.

Creation of synthetic supply chains. By 2027, we predict that the creation of shell companies will be significantly automated through the use of capable AI agents. We pinpoint the key capabilities involved as: (1) *Information retrieval: Automated Incorporation.* An AI agent can navigate the corporate registries of jurisdictions with lax oversight. (2) *Production of Synthetic Documentation.* Generative AI models can produce high-fidelity trade documents, including bills of lading, invoices, end-user certificates, and even fraudulent business materials and online presences, that are likely to bypass manual inspection criteria. These outputs can also be randomized to evade template-matching detection methods. (3) *Cybersecurity: Deepfake Verification.* To bypass remote inspections or KYC checks, networks can deploy real-time deepfake technology to spoof “liveness detection” methods, as generative video models can simulate physiological responses like pulse and blinking.

Agent-assisted network interdiction. A continuous monitoring agent operating on CBP trade data flags a pattern: three ostensibly unrelated resellers in Malaysia have each placed orders for small batches of H200 GPUs from different U.S. distributors within a 10-day window. Individually, each order falls below volume thresholds that would trigger enhanced screening. The monitoring agent cross-references corporate registries and identifies that all three resellers share a registered agent, were incorporated within six weeks of each other, and list directors whose names are transliteration variants of the same two individuals. This example highlights several key capabilities on the enforcement side. They include: (1) *Continuous Monitoring:* Continuous screening of trade data identifies the temporally clustered orders that manual review, processing shipments individually, would not connect. (2) *Entity Resolution:* The agent cross-references ownership records, addresses, and director names across Mandarin and English corporate registries, resolving transliteration variants and mapping the shared registered agent—a task that would require a Mandarin-literate analyst days of manual work. (3) *Administrative Processing:* The agent compiles a structured escalation packet—invoice IDs, payer details, consignee information, entity relationship graph, and risk score—formatted for immediate review by an OEE investigator. The two scenarios are therefore not symmetric. In the circumvention case, AI relaxes bottlenecks close to

current operational limits, while in the enforcement case, AI mostly produces a better lead that still has to survive review, authority, and handoff before it changes the outcome.

5. From AI Capability Gains to Diversion Reduction

A suspicious transaction may be identified but not paused at the point where payment, shipment, export clearance, or remote access can still be stopped. Visibility may weaken once hardware moves through resellers, onward transfers, leasing arrangements, integrated systems, or cloud and service-based channels. And even when one actor has a strong signal, that signal may not reach another actor in a form that still supports timely action.

On the circumvention side, AI addresses tasks closer to current operational limits: generating paperwork, coordinating intermediaries, managing purchases across jurisdictions, and adapting when part of a network is exposed. The result is that AI can improve screening on the enforcement side without producing equally large improvements in actual interruption. At the same time, because enforcement and circumvention adapt to one another, screening itself does not remain fixed. As firms and agencies improve detection, circumvention networks can respond by changing front companies, intermediaries, documentation, and transaction patterns, making risky buyers or end users harder to identify in advance.

5.1. When AI-enabled detection reduces diversion

Enforcement-side AI gains reduce diversion only when they reach a stage at which action is still possible, when visibility persists after the first transfer, and when useful signals move across organizations without rebuilding cases from scratch. First, suspicious cases must reach a point at which someone can still act. Better detection matters only if it arrives at payment release, shipment booking, export clearance, or approval for remote access to controlled compute. If unresolved uncertainty does not trigger pause, escalation, or denial at those points, better detection produces larger alert volume without more interrupted transfers.

Second, visibility has to continue after the first transfer. Diversion often becomes harder to detect when hardware moves through reseller networks, onward transfers, leasing arrangements, or cloud and service-based delivery. In remote compute arrangements, the end user may never take possession of the underlying hardware. The relevant monitoring task is then not paperwork review at the start but retention of structured information about location, control, and effective access. Even strong monitoring tools depend on underlying records (deployment attestations, inventory baselines, serial-number histories, provider-side access logs) that are often incomplete.

Third, useful signals must move across organizations. Exporters, licensing officials, customs authorities, investigators, cloud providers, distributors, and foreign partners each see only part of the pathway. Better screening in one organization matters only if the result can be handed off in a way another organization can use. That requires shared identifiers, basic data fields, clear escalation rules, and evidence that remains understandable when it crosses organizational boundaries.

Fourth, the delivery model of controlled compute must stay within the enforcement target. Export controls were designed around physical transfers of hardware, but advanced AI compute can be accessed through cloud and service-based channels that decouple physical possession from effective use. Entities in restricted jurisdictions can contract for compute services hosted abroad, establish tenant accounts in foreign data centers, and execute AI workloads remotely without any physical transfer occurring. Physical chokepoints such as customs inspections, shipment monitoring, and post-shipment verification do not apply at that layer. The Remote Access Security Act extends export control coverage to controlled compute provided to restricted jurisdictions, and industry has begun deploying GPU location verification, but detecting and proving violations in service-based contexts requires workflows that rely on private-sector compliance and indirect indicators of misuse rather than physical inspection. Residual pathways persist through intermediaries in neutral jurisdictions, compromised legitimate accounts, decentralized compute marketplaces, and third-country providers outside U.S. regulatory reach. AI-enabled detection of physical diversion does not reduce diversion overall if restricted end users can shift to remote compute (see Appendix E).

These conditions are not guaranteed by AI adoption. Agencies and firms can usually deploy AI for document review and case sorting faster than they can redesign approval rules, mandate post-shipment reporting, or build reliable interagency coordination. Government procurement and authorization delays (FedRAMP, FISMA, authority-to-operate) widen this gap further. Circumvention networks face none of these constraints. The likely pattern, absent institutional change, is therefore better screening without proportional improvement in conversion to interdiction. Even the screening advantage should not be treated as fixed. Because enforcement and circumvention adapt to one another, improvements in screening prompt networks to change front companies, documentation, and transaction patterns. Some screening gains will erode, particularly if underlying approval rules and reporting requirements remain unchanged.

5.2. Indicators and implications for enforcement design

Alert volume is a poor measure of enforcement effectiveness because it does not distinguish between three different

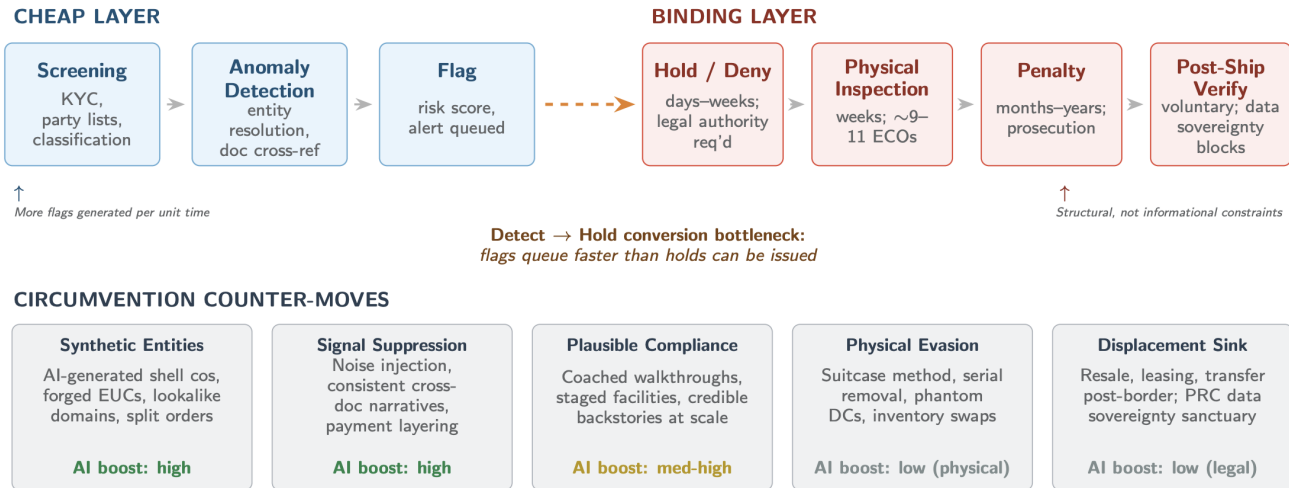


Figure 1. Enforcement pipeline and circumvention counter-moves

outcomes: better screening, stronger interruption, and simple displacement. Three indicators are more diagnostic than alert volume over a 12–18 month window. First, the diversion success rate: whether more suspicious transactions are turned into holds, denials, or seizures rather than being flagged and released. Second, detect-to-hold conversion speed: whether the time from initial flag to binding action shortens. Third, the share of diversion migrating to post-shipment channels, including resale chains, leasing arrangements, and cloud and service-based compute access, which distinguishes genuine reduction from downstream displacement (see Appendix E). A credible claim of enforcement success requires the first two to improve while the third holds steady or declines. A rise in holds or denials is therefore insufficient evidence of success if diversion shifts into reseller chains, leasing arrangements, integrated systems, or remote compute access. The relevant outcome is whether controlled hardware or compute reaches restricted users, not whether the front end of screening becomes more productive.

- (i) Link unresolved uncertainty to mandatory pause or escalation at the points at which a risky transaction can still be stopped (payment release, shipment booking, export clearance, remote compute approval). This attacks the action-stage bottleneck directly by converting flags into holds without requiring new physical inspection capacity.
- (ii) Mandate transfer-event reporting for ownership, location, or effective-control changes after the initial export. This extends post-shipment visibility within existing legal authority.
- (iii) Compress government procurement timelines for enforcement AI tools through faster authorization pathways. This narrows the adoption gap between enforcement and circumvention.
- (iv) Expand physical verification capacity through allied

coordination in major transshipment hubs and through remote verification tools (device identity checks, location verification, facility reconciliation). This is the most institutionally demanding of the four and offers the smallest marginal effect on the detect-to-action gap, because it expands verification rather than conversion.

6. Conclusion

Advanced AI agents will improve enforcement efficiency in targeted domains: administrative processing, anomaly detection, multilingual analysis, and continuous monitoring. These gains concentrate in screening and triage, where throughput can increase substantially. Physical verification, the legal authority to convert detection into interdiction, and post-shipment observability are not constrained by computational capacity, and AI therefore cannot relax them.

On the circumvention side, AI relaxes constraints closer to the binding limits of current operations: coordination, documentation, and planning capacity. Under default adoption trajectories, the enforcement–circumvention balance therefore shifts further toward offense. AI accelerates both sides of the contest without altering its structural conditions, a pattern consistent with the dynamics documented in cybersecurity, where defensive technology improvements are met by offensive adaptation and the long-run balance depends on structural conditions rather than the capabilities of any individual tool (Garfinkel & Dafoe, 2019; Slayton, 2017).

Four structural changes are necessary to shift the equilibrium: linking checks to operational chokepoints, expanding physical verification through allied cooperation, mandating transfer-event reporting, and compressing government procurement timelines. Without these reforms, AI adoption is likely to increase the volume of cases identified without proportionally increasing the volume of cases resolved.

Impact Statement

We analyze how emerging AI capabilities affect the enforcement of U.S. export controls on advanced semiconductors. Our findings have direct implications for national security policy and regulatory design. We do not provide operational guidance for circumvention; the circumvention pathways we describe are drawn from public DOJ prosecution materials and published reporting. The policy recommendations are directed at strengthening enforcement effectiveness within existing legal frameworks.

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Appendix Contents

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A. Definitions and Summary Tables

Table 1. Key terms and definitions.

Term	Definition
Agentic AI	AI systems capable of autonomously planning and executing multi-step actions in pursuit of a goal without continuous human oversight, distinguished from conventional AI tools by their capacity to decompose objectives into subtasks, execute them sequentially across digital environments, and adapt based on intermediate results.
AI-relaxable constraint	A constraint satisfying three conditions: (i) its binding force derives primarily from the cost, speed, or accuracy of information processing; (ii) existing or near-term AI systems demonstrate applicable capability; and (iii) its relaxation requires no changes in legal authority, physical presence, jurisdictional reach, or institutional coordination.
Structurally binding constraint	A constraint that persists regardless of computational capability because its relaxation requires action outside the domain of information processing. Examples include physical verification capacity, legal authority, jurisdictional reach, and government procurement timelines.
Agentic nearcast	A short-horizon analytical exercise adapted from forecasting terminology that maps emerging agentic capabilities onto specific workflows to assess their near-term (2026–2029) differential impact. Distinguished from forecasting by its restriction to capabilities already demonstrated in benchmarks or deployment (Karnofsky, 2022).

B. Constraint Decomposition

This appendix provides the full constraint inventory underlying the informational-vs-structural classification introduced in Section 3.4. Importance weights (high, medium, low) reflect each constraint’s contribution to its side’s overall operational limitation, assigned independently by each co-author based on enforcement-action records and practitioner interviews, with disagreements resolved through discussion.

AI Agents and the Enforcement-Circumvention Balance in Export Controls

Table 2. U.S. export control enforcement architecture for controlled AI chips, illustrating the progression from licensing to post-shipment verification.

Step	Action	Agency	Detail
<i>Stage 1: Licensing & Authorization (Days to weeks)</i>			
1	License application (SNAP-R)	BIS	Product classification, destination, buyer, end-use
2	Initial review	BIS (Commerce)	Completeness, accuracy, risk indicators
3	Interagency review	DoD, DoE, State	Security, foreign policy, proliferation risk
4	Pre-license checks (optional)	BIS Field Office	On-site verification of buyer legitimacy
5	License decision	BIS	Approved, denied, or returned without action
<i>Stage 2: Export Execution (Minutes to hours)</i>			
6	File EEI in AES	Exporter	Exporter identity, buyer, ECCN, license number
7	Outbound targeting & inspection	CBP	Automated screening; physical inspection if flagged
8	Shipment leaves U.S.	—	Goods exported to declared destination
<i>Stage 3: Post-Shipment Enforcement (Weeks to years)</i>			
9	Post-shipment verification	BIS, U.S. Embassy	Site visits confirming item present and as declared
10	Investigation of violations	BIS/OEE	Diversion, smuggling, false documentation
11	Enforcement outcomes	DOJ, BIS	Civil penalties, prosecution, entity listing

Table 3. Enforcement constraints by type and location.

Constraint	Where it appears	Mostly affects	Why it matters
Fragmented data and triage overhead	Screening and case-building	Finding risky cases	Officers spend a lot of their time locating and reconciling data rather than analyzing it
Misclassification and misreporting in filings	Initial export records	Finding risky cases	Visibility depends on exporter-reported descriptions, which can be inaccurate or manipulated
Weak cross-jurisdictional entity resolution	Screening and investigation	Finding risky cases	Shell companies and alias networks are hard to link before they accumulate traceable history
Insufficient physical inspection capacity	Shipment checks and site visits	Verifying and stopping	Better targeting does not create more inspectors on the ground or more site visits
Downstream visibility loss after first export	Post-shipment monitoring	Verifying and stopping	The system sees the first hop but often loses sight of later transfers where diversion occurs
Weak chain-of-custody and traceability	Resale, integration, deployment	Verifying and stopping	Suspicion can rise faster than the government can verify where hardware actually is
Jurisdictional barriers to evidence	Overseas verification and investigation	Verifying and stopping	Legal access to records often weakens or disappears once goods leave U.S. territory
Interagency coordination and handoff delays	Escalation and enforcement action	Verifying and stopping	Cases must move across institutions that operate on different timelines and procedural requirements

B.1. Enforcement-Side Constraints

Table 4. Enforcement-side constraint decomposition with rationale.

ID	Constraint	Class.	Imp.	Rationale
E1	Data fragmentation and triage overhead	AI-relaxable	High	Enforcement and compliance officers spend roughly 80 percent of their time locating and reconciling data across siloed systems rather than analyzing it. AI can compress this overhead by roughly $2\text{--}3\times$.
E2	Limited capacity for cross-jurisdictional entity resolution	AI-relaxable	High	Matching company names, addresses, aliases, and transliterations across registries and languages is an information-processing task where deployed systems (Sayari, Altana) and GNN-based approaches demonstrate strong capability.
E3	Limited multilingual document analysis capacity	AI-relaxable	Medium	The small number of Mandarin-literate analysts constrains cross-language entity matching and document review. Frontier language models perform well on translation and cross-language analysis, but the enforcement benefit depends on integration with government data systems.
E4	Weak detection of misclassified or misdeclared exports	AI-relaxable	Medium	AI pattern detection can flag inconsistent item classifications in trade data, but cannot independently verify the physical contents of shipments or the accuracy of exporter declarations. The informational component is relaxable; the verification component is not.
E5	Insufficient physical inspection and verification capacity	Struct. binding	High	BIS maintains approximately 150 domestic investigators and 9–11 overseas Export Control Officers, conducting on the order of 100 Pre-License Checks per year (practitioner interviews 2025–2026). Inspection requires physical presence that computation cannot substitute.
E6	Legal authority and evidentiary requirements for enforcement actions	Struct. binding	High	Shipment holds, license denials, seizures, Entity List designations, and criminal prosecutions each require legal authorization and evidence meeting procedural standards. These requirements define the rate at which flags can be converted into enforcement outcomes, and cannot be satisfied computationally.
E7	Jurisdictional barriers to evidence collection	Struct. binding	High	China’s data sovereignty laws (Data Security Law, Counter-Espionage Law) and the voluntary nature of overseas end-use checks limit access to records once goods leave U.S. territory. The constraint is legal access to evidence, not analytical capacity.
E8	Loss of visibility after initial export	Mixed	High	Monitoring follows a visibility gradient: strong at the initial U.S. export, weak at the first foreign resale, often nonexistent by the second or third transfer. AI can extend monitoring within existing data streams, but cannot create visibility where records do not exist or where legal barriers prevent access.
E9	Interagency coordination and handoff delays	Struct. binding	Med.	Enforcement requires handoffs across licensing (BIS), customs (CBP), investigation (OEE), and prosecution (DOJ). Coordination timelines are constrained by institutional procedures and interagency protocols, not by information-processing speed.
E10	Government AI procurement timelines	Struct. binding	Med.	FedRAMP, FISMA, and authority-to-operate requirements impose procurement cycles of 18–36 months. These timelines are institutional, not computational, and create a persistent adoption lag relative to commercially available tools.

B.2. Circumvention-Side Constraints

Table 5. Circumvention-side constraint decomposition with rationale.

ID	Constraint	Class.	Imp.	Rationale
C1	Maintaining documentation coherence at scale	AI-relaxable	High	Internal consistency across shell companies, trade documents, corporate records, and end-user certificates grows complex as network size increases. Errors or inconsistencies propagate, raising detection risk. This is an information-processing challenge that generative AI directly addresses.
C2	Multi-step coordination across intermediaries and jurisdictions	AI-relaxable	High	Synchronizing logistics, payment timing, and shipping across multiple actors and jurisdictions requires multi-step planning. Agentic AI systems demonstrate growing capability to decompose and execute such workflows (Kwa et al., 2025).
C3	Speed and cost of shell company creation and incorporation	AI-relaxable	High	Navigating corporate registry interfaces, preparing jurisdiction-specific documentation, and maintaining consistency across filings currently require human operators to execute manually across each jurisdiction. These are automatable workflows.
C4	Dependence on trusted human intermediaries at custody transfer points	Struct. binding	High	Procurement agents, warehouse operators, and logistics handlers face asymmetric personal risk, including criminal prosecution and imprisonment. Trust relationships at points of physical custody cannot be automated, and network expansion increases the number of exposed participants.
C5	Financial routing through regulated payment systems	Struct. binding	High	Payments for high-value semiconductor transactions must traverse regulated financial systems. Layering transactions increases complexity and latency, while mismatches between declared and actual counterparties remain a recurring point of detection (Department of Justice, 2025).
C6	Counter-detection and operational security	AI-relaxable	Medium	Monitoring enforcement patterns, Entity List updates, and investigative indicators via OSINT platforms is an information-processing task. This capability is defensive and reactive: it reduces exposure but does not directly increase circumvention throughput.
C7	Bypassing remote verification and KYC checks	AI-relaxable	Medium	Deepfake identity documents and staged remote inspections are enabled by generative video and document models, but detection methods are also advancing. The constraint is partially relaxable, with the net effect depending on the screening-evasion arms race.
C8	Network reconstitution speed after enforcement disruption	AI-relaxable	Medium	Dissolving and recreating entities, rerouting logistics, and establishing new commercial histories can be accelerated by AI, but each cycle of reconstitution still temporarily degrades trust relationships and reduces operational throughput.
C9	Physical handling, storage, and transport of controlled items	Struct. binding	Low	Irreducibly dependent on human operators, but the constraining force is limited for small, high-value items like individual GPU modules, which are easy to conceal and transport. Physical logistics function as an operational inconvenience rather than a binding limit on throughput.

B.3. Aggregate Assessment

Available evidence suggests AI can increase screening and triage throughput by a factor of two to three in the near term. Dell’Acqua et al. (2023) found AI-augmented BCG consultants completed tasks 25 percent faster with 40 percent higher quality on tasks within AI’s capability frontier, though performance degraded on tasks outside that frontier. McKinsey & Company (2025) reports 15 to 20 percent productivity gains from pre-agentic AI in AML/KYC investigation, with projections of larger gains as agentic architectures mature. Allen (2025) estimated that modern AI/ML technologies could yield five- to tenfold greater analyst productivity in export control enforcement specifically, though this estimate has not been empirically validated. We use the conservative end of the available evidence—roughly two to three times for screening and triage tasks—while noting that higher gains are plausible for narrow subtasks such as document sorting or name matching, where near-full automation is feasible.

The stages where enforcement must convert flags into holds, denials, inspections, or penalties scale far more slowly, because they require legal authority, physical presence, and institutional coordination. We estimate these stages improve at roughly 1 to 1.5 times current throughput under AI augmentation, reflecting the judgment that computational tools can modestly accelerate case preparation and coordination but cannot substitute for the non-informational requirements that define these stages. The 5–10× figure reported in Allen (2025) and in Figure 1 applies to narrow, highly automatable subtasks such as document sorting and name matching, where near-full automation is feasible. The 2–3× estimate applies to combined screening and triage pipelines where human review remains in the loop. The detect-to-action gap therefore does not depend on a precise throughput estimate. The core claim is comparative: across plausible screening-side gains, the stages that convert flags into holds, denials, inspections, or penalties remain limited by legal authority, physical presence, and institutional coordination, so detection capacity can expand faster than enforceable action.

Weighting the constraint inventory by importance yields the following approximate ranges. On the enforcement side, 30–50 percent of binding constraints are AI-relaxable and 50–70 percent are structurally binding. On the circumvention side, 55–75 percent of binding constraints are AI-relaxable and 25–45 percent are structurally binding. The ranges are wide because the weighting is qualitative, the inventory may not be exhaustive, and the co-authors did not assign identical importance weights.

The directional conclusion does not depend on the specific ranges. It would reverse only if enforcement’s AI-relaxable share exceeded circumvention’s, which would require reclassifying physical verification capacity or legal authority as primarily informational problems, or reclassifying documentation coherence and multi-step coordination as structurally binding. Physical verification requires physical presence by definition; documentation coherence is an information-processing task by definition. Neither reclassification is defensible given the operational evidence in Sections 2.1 and 3. The qualitative conclusion holds for any reader who accepts the directional asymmetry, regardless of the specific ranges. The conclusion is also stable under reasonable variation in individual weights: upgrading interagency coordination from medium to high importance on the enforcement side raises the AI-relaxable share to approximately 35–55 percent, still below the circumvention range. The ranges overlap only under simultaneous reclassification of multiple structurally binding enforcement constraints as AI-relaxable.

C. Agentic Capability Assessment

C.1. Methodology and Capability Table

We restrict the scope of the capability analysis to agentic capabilities that (i) map to specific tasks documented in BIS enforcement actions, DOJ prosecutions, or think-tank analyses, and (ii) we anticipate will change rapidly during the 2026–2028 window and remain a priority for frontier labs, inferred from benchmarks, expert forecasts, and broader deployment signals.

Each capability is scored on two confidence axes. Growth measures confidence that the capability will have significantly evolved by 2028. Significance measures confidence that the mature capability can meaningfully impact the current enforcement–circumvention dynamic. We constrain the confidence range by the quality of available evidence: 75 percent and higher (Tier 1) corresponds to high-signal data from multiple expert or lab forecasts, benchmark trends, and deployed systems; 50–75 percent (Tier 2) corresponds to some expert views and benchmarks or direct deployment evidence; 50 percent and lower (Tier 3) primarily involves trend extrapolation, market signals, and speculative judgment. The five-point scoring scale ranges from 5 (Decisive, multiple independent evidence lines converging) to 1 (Minimal, speculative with no direct evidence). Net impact reflects the direction of tilt under default adoption trajectories.

The seven capability assessments split four-to-three in favor of enforcement, but the count is misleading. Circumvention’s advantage is driven by early adoption and larger error bandwidth, not by the number of favorable capabilities. What matters

Table 6. Agentic capability assessment. *Growth* measures confidence that the capability significantly evolves by 2028. *Significance* measures confidence that the mature capability meaningfully shifts the enforcement–circumvention balance. *Net Impact* reflects the direction of tilt under default adoption trajectories. Scores follow the five-point scale defined in the text.

Capability	Growth	Signif.	Net Impact	Rationale
Administrative Processing & Record-Keeping	5	3	Enforcement	Network-level document analysis accessible only to entities with comprehensive trade data (CBP, Altana) outweighs circumvention’s generative document gains over time.
Task Decomposition	4	3	Circumvention	Circumvention deploys faster, tolerates higher error rates, and has more manual workflows to automate.
Evidence Collection & Information Retrieval	5	4	Enforcement	Government access to protected databases and classified records creates a structural advantage in entity resolution and anomaly detection.
Autonomous Engineering & Planning	5	4	Circumvention	Circumvention faces no procurement barriers, tolerates failure, and plans against a slow, observable opponent.
Cybersecurity & Adversarial Robustness	5/3	5	Circumvention	Fundamental attacker-defender asymmetry: offense needs one successful breach, defense must hold at ~100%. Enforcement systems process untrusted external data, amplifying the attack surface.
Long-Horizon Monitoring	4	3	Enforcement	Structurally an enforcement capability: requires persistent access to comprehensive data streams only government agencies possess.
Multilingual Comprehension	5	3	Enforcement	Directly addresses enforcement’s binding language-barrier constraint, while circumvention networks are already natively bilingual.

is whether the capabilities relieve constraints that are actually binding, not how the gross distribution of new tools divides between the two sides.

C.2. In-Depth Discussion of Three Major Capabilities

We now examine in detail the three capabilities where the asymmetries are most pronounced: evidence collection, autonomous engineering and planning, and cybersecurity.

C.2.1. EVIDENCE COLLECTION AND INFORMATION RETRIEVAL

Agentic open-source intelligence (OSINT) collection is already deployed at scale: Sayari processes 5.3B trade records across 100+ risk categories, and Altana AI maintains a knowledge graph of 500+ million corporate entities with trade relationship mapping. By leveraging ML in processing check images and metadata continuously, the U.S. Treasury recovered \$1B in check fraud in FY2024 alone. Graph Neural Networks (GNNs) achieve robust entity resolution even with limited labeled data, and companies like Uber, Google, and Alibaba have shifted to GNN-based approaches for network analysis.

Enforcement benefits from automated entity resolution, with agents cross-referencing corporate registries, names, addresses, and transliterations across languages and jurisdictions. Enforcement agencies also have access to comprehensive government-protected databases: customs data, ownership records, and multilateral information channels. Accelerated discrepancy and anomaly detection can help flag patterns that circumvention actors cannot foresee.

Circumvention actors could use the same tools for counter-intelligence—identifying which shell companies have been flagged, mapping enforcement investigative patterns, and monitoring exposure via OSINT platforms. This benefit is defensive and reactive: circumvention can detect exposure only after being flagged, and cannot prevent the overall tightening

of the enforcement net.

Net Impact: Favors enforcement. Evidence collection and anomaly detection favor the side with legitimate access to government databases and classified records. This advantage rests on modernizing BIS infrastructure; if modernization is delayed, the enforcement benefit remains largely unrealized and the net impact narrows toward parity.

Capability Forecast: [Growth: 5, Significance: 4]. By 2028, entity resolution should incorporate real-time cross-language capabilities: mapping Chinese company names and variants from Tianyancha/NECIPS against the BIS Entity List and the OFAC SDN list, producing documented shell company networks. GNN-based entity resolution combined with knowledge graphs (e.g., Altana) can enable multi-tier supply chain tracing. Impact confidence remains moderate due to dependency on government data infrastructure modernization. In 2024, CSIS reported that BIS databases remain in “extremely poor shape.” We anticipate this to change by 2028 due to growing awareness and enforcement pressure.

C.2.2. AUTONOMOUS ENGINEERING AND PLANNING

AI agents can autonomously write, test, and deploy software and decompose complex objectives into sequential subtasks executed across digital environments. These are two facets of the same underlying advance: the extension of reliable autonomous task horizons from minutes to hours to days. In mid-2025, frontier AI models resolved over 70 percent of real software engineering tasks on SWE-Bench Verified, up from approximately 20 percent in 2024 (Scale AI, 2026). METR benchmarks show that the time horizon of reliably completed tasks doubles approximately every seven months, accelerating to every four months in 2024–2025 (Kwa et al., 2025). Frontier models achieve comparable performance with classical planners on standard PDDL benchmarks (Corrêa et al., 2025). A complexity bottleneck persists: SWE-Bench Pro scores remain below 25 percent for enterprise-level tasks, and plan quality degrades substantially with horizon length, with errors compounding as multi-session execution extends across days or weeks. AI 2027 scenario analysis predicts “superhuman coders” by March 2027, though this is disputed in the broader AI community (Kokotajlo et al., 2025). Amodei (2025) has claimed that most code at Anthropic is now agent-written, though independent analysis puts the company-wide figure closer to 50 percent when measured by merged code rather than initial drafts (Greenblatt, 2025). Other major coding assistants (Copilot, Cursor, Codex) are driving rapid adoption across the industry.

Enforcement benefits from agent-assisted pre-screening of end-use checks, automated case-management systems, and custom anomaly-detection pipelines. Agents can scope corporate networks through alias and transliteration discovery, map beneficial ownership relationships, and build risk profiles. Enforcement, however, requires enterprise-grade reliability and auditability, bottlenecking deployment on technical implementation, procurement timelines, and inter-agency coordination. False positives impose costs (delayed legitimate shipments, diplomatic friction), and long-horizon investigation plans require extensive human oversight (legal review, supervisory approval, congressional accountability).

Circumvention actors benefit from accelerated multi-step execution across company registration, document generation, transshipment logistics, and payment coordination. Circumvention faces fewer barriers to deployment: no procurement cycles, no auditability requirements, and readily available commercial infrastructure. Circumvention tolerates higher error rates—if one shell company setup fails, another can be attempted at minimal loss. Circumvention also plans against a known, slow, resource-constrained opponent whose enforcement patterns are partially observable.

Net Impact: Favors circumvention. The asymmetry is driven by three factors: (1) adoption speed, as circumvention deploys immediately while government is bottlenecked by procurement clearances; (2) error tolerance, as circumvention can deploy earlier models since failures are more permissible; (3) planning complexity, as enforcement plans against an opaque, adaptive adversary while circumvention plans against a slow, observable one. The advantage is constrained by execution bottlenecks at longer horizons, and enforcement has potential to leapfrog via contracts like CBP–Altana.

Capability Forecast: [Growth: 5, Significance: 4]. If the 4-month doubling rate holds, the 50-minute task horizon extends to multi-hour tasks by mid-2026 and multi-day tasks by end-2027 (Metaculus, 2025). Software engineering capabilities will continue improving, though whether autonomous agents can support enterprise-grade complexity remains the key uncertainty. Plan quality degrades non-linearly with horizon length, and persistent memory architectures, while improving, face unresolved challenges in multi-month strategy execution. Impact confidence is moderate because enforcement’s more complex systems take longer to build and deploy, while circumvention needs only simpler, readily available tools.

C.2.3. CYBERSECURITY AND ADVERSARIAL ROBUSTNESS

Agentic impact on cybersecurity remains actively debated. In 2025, OWASP ranked prompt injection as the #1 critical LLM vulnerability. Systematic red-teaming evaluations show attack success rates of 87.2 percent for GPT-4 and 82.5 percent for

Claude 2, with jailbreak prompts transferring across model families at rates up to 64.1 percent (Pathade, 2025). A joint paper by researchers from AISI, Anthropic, and the University of Oxford showed that 12 published defenses against prompt injection were bypassed with attack success rates above 90 percent (Souly et al., 2025). Commercially, Deloitte projects AI-enabled fraud losses in the U.S. to reach \$40B by 2027.

Enforcement. Whether enforcement agencies deploy agents for screening or investigation, those agents become targets for adversarial manipulation. Any enforcement AI that processes external documents (shipping records, corporate filings, trade data) is vulnerable, requiring human review at every decision point, as circumvention actors can craft prompt injections specifically designed to cause enforcement AI to misclassify risky shipments as benign. The most likely attack vectors by 2028 include prompt injection in document analysis AI, deepfake identity documents undermining KYC verification, evasion of anomaly detection, and direct manipulation of enforcement agent workflows.

Circumvention. Circumvention actors will deploy advanced agents to reveal and target vulnerabilities, jailbreaking models to generate fraudulent documentation, impersonation content, or operational planning assistance. As circumvention systems primarily process trusted internal data (their own logistics, communications, financial flows), they face far less adversarial exposure. The more serious risk emerges from state-backed networks with access to frontier technical capabilities.

Net Impact: Favors circumvention. The asymmetry is driven by three factors: (1) the attacker-defender asymmetry is fundamental, as offense needs one successful jailbreak while defense must remain at ~100 percent; (2) enforcement systems have more attack surface as they must process untrusted external data; (3) the likely evolution of state-backed adversaries will amplify the resources available for developing more sophisticated attacks.

Capability Forecast: [Growth: 5 (offense), 3 (defense), Significance: 5]. Cybersecurity capabilities are dual-use, with historic advantage on the offensive side. Reports presented at NeurIPS 2024–25 and evidence published by AISI indicate that adversarial ML researchers are generally pessimistic about closing the gap by 2028. The fundamental asymmetry—attacks are cheap to generate, defenses are expensive to verify—follows traditional cybersecurity patterns but will likely worsen as the attack surface is essentially unbounded.

C.3. Detect–Action Gap

The asymmetry across the seven capabilities is captured most cleanly in the *detect–action gap*. AI-enhanced screening can increase flag generation by roughly $2\text{--}3\times$ in combined pipelines, with higher gains on narrow subtasks, while institutional mechanisms for converting flags into binding enforcement actions scale at roughly $1\text{--}1.5\times$. The gap between detection capacity and action capacity is the structural feature that tilts the balance toward circumvention under default adoption trajectories. Closing it requires enforcement to (i) press its structural advantages in data access and legal authority before circumvention adapts, and (ii) resolve systematic bottlenecks: modernization of BIS infrastructure, early deployment of frontier technologies, and procurement timelines that run in institutional rather than capability time.

D. Workflow Mapping: Defense–Offense Paired Functions

The following table maps each enforcement workflow to its circumvention counterpart and identifies the binding constraint governing that pair. The mapping makes explicit that AI improvements on one side are met with adaptive counter-moves on the other, and that the binding constraints at each pairing are typically structural rather than informational.

E. Cloud and Service-Based Circumvention

This appendix examines a complementary circumvention vector: the provision of advanced AI compute through cloud and service-based delivery models. The pattern illustrates how delivery-model shifts compound the structural advantages identified in the main analysis. When physical export of restricted AI chips is constrained, access to compute can be obtained remotely by leasing time on foreign-located servers equipped with controlled hardware, decoupling physical possession from effective use.

E.1. Characteristics of Cloud-Based Access

Export control regimes have historically treated the transfer of physical items as the central object of regulation. Cloud-based compute complicates this framework by allowing the functional benefits of controlled hardware to be accessed without a transfer of possession. Under earlier regulatory interpretations, the provision of compute time or remote access to hardware was not unambiguously classified as an export of the underlying technology. Entities located in restricted jurisdictions could legally contract for compute services hosted abroad, establish tenant accounts in foreign data centers, and execute AI

Table 7. Enforcement–circumvention workflow mapping with binding constraints.

Defense Workflow	Offense Counterpart	Binding Constraint
<i>Verification:</i> validating identity, end-use, and documentation claims (e.g., identity checks, KYC).	<i>Fabrication:</i> creation and curation of fake identities, records, and traces (e.g., forged end-user certifications, shell company incorporation, physical facility imitation).	Physical verification capacity; fabrication quality detectable only at scale through graph-based analysis.
<i>Monitoring:</i> continuous tracking and anomaly detection over time (e.g., chip shipment tracing).	<i>Signal suppression:</i> degrading observability through telemetry poisoning and noisy signals (e.g., routing through intermediaries, inconsistent descriptions, fake payment trails).	Data infrastructure modernization; monitoring is only effective when coupled to binding follow-through.
<i>Investigation:</i> tracking active cases and building profiles by systematizing data (e.g., case-building, circumvention pathway tracing).	<i>Deception:</i> evidence destruction or engineering to mislead analysis and reduce visibility (e.g., destroying audit trails, false-information floods).	Legal authority and inter-agency coordination; investigation timelines measured in months vs. circumvention cycles measured in days.
<i>Pattern Detection:</i> analyzing signals and linking records to compose the “true” picture of actors (e.g., identifying clustering, recording entities and profiles).	<i>Segmentation:</i> promoting record segmentation to prevent entity resolution (e.g., alternating shells and directors, splitting activities across hubs).	Data integration across jurisdictions; segmentation exploits the fragmentation AI alone cannot resolve.
<i>Interdiction/Neutralization:</i> applying privilege escalations and enforcement task chains to block or contain diversion.	<i>Capture:</i> shifting defender decisions by compromising system controls (e.g., insider recruitment, corruption of auditors, registries, and licensing points).	Legal authority and jurisdictional reach represent the ultimate binding constraint that no AI capability can substitute.
<i>Guarding:</i> cyberdefense and system integrity, hardening defenses against evasion techniques.	<i>Disruption:</i> active system attacks to disable cyberdefenses (e.g., jailbreaks, cyberattacks, DDoS).	Fundamental attacker-defender asymmetry; defense costs scale with attack surface.

workloads remotely. Once compute access is abstracted into a service, physical chokepoints such as customs inspections and shipment monitoring no longer apply, and enforcement shifts toward contract oversight, user attribution, and network-level monitoring.

E.2. Service-Based Circumvention Modalities

Cloud-based circumvention occurs through several distinct service layers. Infrastructure-as-a-Service and Hardware-as-a-Service platforms allow users to rent dedicated GPU capacity or bare-metal access to accelerators hosted in foreign data centers. Higher-level services further abstract the underlying hardware: Software-as-a-Service AI tools perform computation on provider-managed servers, while API-based access to large models allows users to benefit from training and inference conducted entirely on controlled hardware. In these cases, the end user may never directly interact with GPUs, complicating attribution.

E.3. Regulatory Response

U.S. policymakers have moved to address these pathways. The Remote Access Security Act, passed by the House in January 2026, extends export control coverage to the provision of controlled compute resources to restricted jurisdictions. Complementary executive actions emphasize enhanced end-use monitoring and location verification. Industry responses have included software-based GPU location verification mechanisms. These measures collectively aim to align legal definitions with the operational realities of networked compute delivery. Closing the legal gap does not resolve the enforcement challenge: detecting and proving violations in cloud environments requires new workflows that rely heavily on private-sector compliance and indirect indicators of misuse.

E.4. Residual Evasion and Structural Limits

Even under expanded legal authority and improved provider compliance, residual pathways are likely to persist. Adversaries can employ intermediaries in neutral jurisdictions, compromise legitimate cloud accounts, or distribute workloads across many lower-tier instances to avoid detection thresholds. Emerging decentralized compute marketplaces, which match demand and supply for GPU cycles on a peer-to-peer basis, complicate enforcement further. Third-country mediation is a

particularly persistent vulnerability: providers operating outside U.S. or allied regulatory reach may offer compute services using gray-market hardware. While extraterritorial tools such as the Foreign Direct Product Rule can be applied in principle, identifying violations and sustaining enforcement pressure in service-based contexts is more complex than for physical exports.

E.5. Assessment

Measures targeting cloud-based circumvention raise the cost and complexity of accessing restricted AI compute but do not eliminate access. Enforcement effectiveness depends on sustained provider cooperation, allied alignment, and the ability to attribute service usage to ultimate beneficiaries. As delivery shifts further toward services and abstraction layers, the offense retains structural advantages in speed, adaptability, and deniability, while enforcement remains constrained by jurisdictional boundaries and limited observability. This pattern—enforcement gains that raise costs without closing pathways—mirrors the broader dynamic identified in the main analysis and reinforces the conclusion that structural measures, not screening improvements alone, are necessary to shift the equilibrium.

F. Data Sovereignty as Defensive Capability

A significant defensive capability on the circumvention side is the legal framework within the PRC. China’s data sovereignty regime—comprising the Data Security Law, the Personal Information Protection Law, the Counter-Espionage Law (as amended in 2023), and the State Secrets Law (as amended in 2024)—creates a legal sanctuary for diverted goods once they cross the border. These laws collectively prohibit providing data to foreign enforcement authorities without government approval and impose severe penalties for violations, ranging from administrative fines to criminal prosecution in cases involving national security ([National People’s Congress, 2021](#)). The practical effect is to prevent multinational corporations from complying with U.S. subpoenas or conducting internal investigations within their Chinese subsidiaries, forcing U.S. enforcement agencies to rely on external indicators and open-source intelligence, which is inherently less reliable than internal corporate data. This asymmetry in information access is a persistent feature of the enforcement landscape that no AI capability can directly overcome.

G. Limitations, Methodology, and Disconfirmation

G.1. Methodological Scope and Evidence Base

This analysis is informed by practitioner interviews, public enforcement records, benchmark data, and the analytical framework developed in the main text, rather than by classified intelligence or comprehensive empirical data on circumvention volumes. The confidence scores in Table 6, the bottleneck composition estimates, and the scenario net-advantage scores were produced through independent scoring by each co-author followed by consensus discussion, informed by both documentary evidence and expert input. We report these as structured qualitative judgments rather than empirically calibrated measurements, and specify disconfirmation criteria in Section G.4. Interviewees included former BIS officials with direct enforcement experience, compliance professionals at major semiconductor distributors, and researchers working on adjacent problems in AI governance and semiconductor policy. We do not name individual interviewees but characterize their institutional positions where relevant to the weight of specific claims.

G.2. Known Uncertainties

Several important questions remain unresolved. No official U.S. government estimate exists for the percentage of export control violations that are detected, though practitioners generally believe the figure to be low given the volume of transactions relative to enforcement capacity. No comprehensive assessment measures whether prosecutions reduce smuggling attempts (the deterrence question). The timeline for BIS deployment of modern AI capabilities remains unclear. The trajectory of Chinese state involvement—specifically when and to what extent state resources might be deployed to professionalize and coordinate smuggling operations beyond current levels—is uncertain; if the Chinese state actively coordinates illicit procurement beyond providing funding, the offense could become substantially more formidable. No comprehensive estimate exists of industry spending on export control compliance.

The scenarios in Section 5 assume that circumvention actors adopt agentic capabilities at uniformly high levels. In practice, adoption is likely to be uneven across networks: some may lack the technical sophistication to deploy frontier AI tools effectively, and others may face constraints on model access if frontier model providers implement effective use restrictions. The assessment is therefore best read as describing the upper bound of the circumvention advantage under widespread

adoption, with actual outcomes depending on heterogeneity that is difficult to forecast. If frontier model providers restrict access for circumvention-related tasks through safety filters, use policies, and monitoring, the capability landscape shifts, but smaller open-weight models with fewer restrictions are widely available and fine-tuning can circumvent many safety measures. We treat model access restrictions as a friction that slows but does not prevent circumvention-side adoption.

The analysis also assumes that controlled chips remain primarily available through U.S. and allied supply chains. If Chinese domestic chip production (e.g., SMIC or successors) reaches near-frontier capability levels, the enforcement–circumvention contest changes character, as the demand for diverted chips would diminish. We treat this as a boundary condition rather than a near-term likelihood, but acknowledge it as a structural uncertainty.

G.3. Risk of Agentic Enforcement Deployment

Deploying agentic systems for enforcement introduces new attack surfaces. Graph poisoning—in which adversaries inject fabricated entities or relationships into trade-network databases to deceive graph-based detection systems—becomes more feasible as generative AI reduces the cost of producing convincing forgeries. Adversaries could generate synthetic shell companies with rich digital backstories, though such entities often lack the deep historical graph density that accumulates over years of tax filings, court records, employee churn, and vendor relationships.

G.4. Disconfirmation Criteria

Our central conclusion—that the default trajectory favors circumvention—would be disconfirmed by evidence of sustained increases in blocked controlled volume pre-shipment without a corresponding rise in post-shipment transfer leakage. In the scoring framework, this would mean detect-to-hold conversion rising while the share of diversion shifting to post-shipment channels does not. If this pattern were observed over a 12- to 18-month period, it would indicate that AI-enhanced enforcement is producing net reductions in diversion rather than merely displacing it, and our assessment would require revision.